

Hypothesis Testing: Comparing 2 Populations

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Topics

- Comparing two population means
 - Matched (paired) samples
 - Independent samples
- Comparing two proportions

Questions on A/B test are common in DS interview
<https://datalemur.com/blog/ab-testing-interview-questions-and-answers>

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Testing 2 means Paired vs. Unpaired Test

Suppose that a taste test of two flavors is carried out.

- Paired Test (Matched Samples): Select a simple random sample and let every person in the sample rate each of the two flavors in a random order.
- Unpaired Test (Independent Samples): Select two simple random samples and let each group rate only one flavor.
- Which one is better?

Paired test tends to remove much of the extraneous variations (e.g., variation in people and experimental conditions)



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Difference Between the Means of Two Populations: Matched Samples

- With a matched-sample design each sampled item provides a pair of data values.
- It's feasible for
 - Before and after measurements
 - Cross-over designs (experience both treatments)
- By considering the difference of each pair of the data values, we can apply the same procedures as the single population case.

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To test the delivery times of two express delivery services, UPX (United Parcel Express) and INTEX (International Express), two reports were sent to a random sample of ten district offices with one report carried by UPX and the other report carried by INTEX. Assume delivery times are normally distributed. Do the data indicate a difference in mean delivery times for the two services at 95% confidence level?

Delivery Time (Hours)

District Office	UPX	INTEX	Difference
Seattle	32	25	7
Los Angeles	30	24	6
Boston	19	15	4
Cleveland	16	15	1
New York	15	13	2
Houston	18	15	3
Atlanta	14	15	-1
St. Louis	10	8	2
Milwaukee	7	9	-2
Denver	16	11	5

- We want to test if there is a difference in mean delivery times for the two services
- Let μ_1 = mean delivery times for UPX
 μ_2 = mean delivery times for INTEX

- Hypotheses

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_a: \mu_1 - \mu_2 \neq 0$$

	Unknown σ_1, σ_2 MATCH SAMPLE t-test statistic= $\frac{\bar{d} - D_0}{s_d / \sqrt{n}}$ df= n-1
Upper tailed test $H_0: \mu_1 - \mu_2 \leq D_0$ $H_A: \mu_1 - \mu_2 > D_0$	Critical value = $t_{\alpha, df}$ Reject H_0 if $t \geq t_{\alpha, df}$ p-value= $\Pr(T > t)$
Lower tailed test $H_0: \mu_1 - \mu_2 \geq D_0$ $H_A: \mu_1 - \mu_2 < D_0$	Critical value = $-t_{\alpha, df}$ Reject H_0 if $t \leq -t_{\alpha, df}$ p-value= $\Pr(T < t)$
Two Tailed test $H_0: \mu_1 - \mu_2 = D_0$ $H_A: \mu_1 - \mu_2 \neq D_0$	Reject H_0 if either $t \geq t_{\alpha/2, df}$ or $t \leq -t_{\alpha/2, df}$ p-value= $2 * \Pr(T > t)$

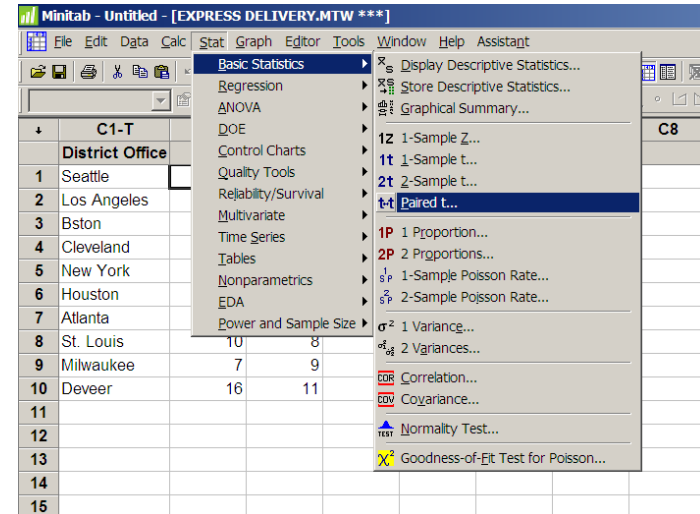
Example: Express Deliveries

$$H_0: \mu_1 - \mu_2 = 0$$

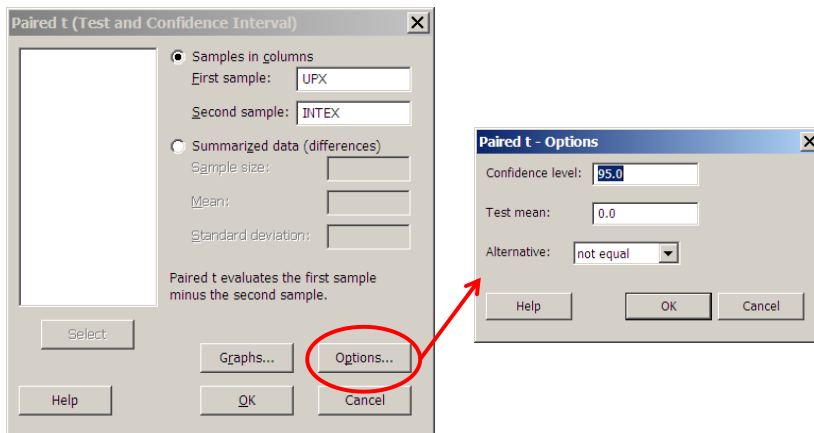
$$H_a: \mu_1 - \mu_2 \neq 0$$

- Sample mean = 2.7; sample standard deviation = 2.908
So t-test statistic =
- Critical value at 95% confidence level =
- Conclusion:

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Select Columns and Set Options



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Paired T-Test and CI: UPX, INTEX

Paired T for UPX - INTEX

	N	Mean	StDev	SE Mean
UPX	10	17.70	7.87	2.49
INTEX	10	15.00	5.64	1.78
Difference	10	2.700	2.908	0.920

95% CI for mean difference: (0.620, 4.780)

T-Test of mean difference = 0 (vs not = 0):

T-Value = 2.94 P-Value = 0.017

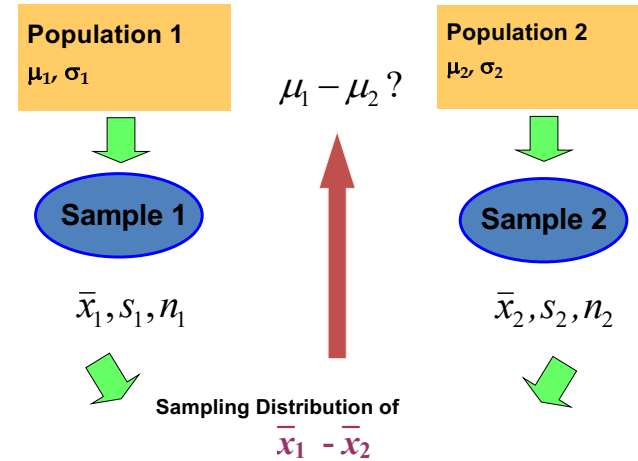
Since p-value = _____ there is _____ to believe that the mean delivery times of the services are different.

Paired Tests May not Always Be Feasible

- Evaluating Marketing Campaign Effectiveness for Different Ads
- Comparing Product Sales in Different Regions
- Comparing the Effect of a Drug on Two Different Groups (treatment v.s placebo)
- Comparing Salaries Between Two Companies

→ Unpaired Test (Independent Samples)

Difference Between the Means of Two Populations: **Independent Samples**



Derive test statistic

We want to test:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_a : \mu_1 \neq \mu_2 \text{ (or one-sided)}$$

1. Sampling distributions of the individual sample means

Assuming each population is approximately normal (or n_1, n_2 large):

$$\bar{X}_1 \sim N\left(\mu_1, \frac{\sigma_1}{\sqrt{n_1}}\right), \quad \bar{X}_2 \sim N\left(\mu_2, \frac{\sigma_2}{\sqrt{n_2}}\right)$$

where σ_1 and σ_2 are the population standard deviations.

Since the samples are independent, the difference:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right)$$

Comparing two population means

	Known σ_1, σ_2 Independent Two Sample Z-Test z-test statistics=	Unknown σ_1, σ_2 Matched Sample T-Test t-test statistic=	Unknown σ_1, σ_2 Independent Two Sample T-Test t-test statistic =
	$\frac{\bar{X}_1 - \bar{X}_2 - D_0}{\sqrt{\sigma_1^2 / n_1 + \sigma_2^2 / n_2}}$	$\frac{\bar{d} - D_0}{s_d / \sqrt{n}}$ df = n-1	$\frac{\bar{X}_1 - \bar{X}_2 - D_0}{\sqrt{s_1^2 / n_1 + s_2^2 / n_2}}$ df = $\frac{(s_1^2 / n_1 + s_2^2 / n_2)^2}{\frac{1}{n_1 - 1} \left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1} \left(\frac{s_2^2}{n_2}\right)^2}$
Upper tailed test $H_0 : \mu_1 - \mu_2 \leq D_0$ $H_A : \mu_1 - \mu_2 > D_0$	Critical value = z_α Reject H_0 if $z \geq z_\alpha$ p-value = $\Pr(Z > z)$	Critical value = $t_{\alpha, df}$ Reject H_0 if $t \geq t_{\alpha, df}$ p-value = $\Pr(T > t)$	This df computation not required in this class
Lower tailed test $H_0 : \mu_1 - \mu_2 \geq D_0$ $H_A : \mu_1 - \mu_2 < D_0$	Critical value = $-z_\alpha$ Reject H_0 if $z \leq -z_\alpha$ p-value = $\Pr(Z < z)$	Critical value = $-t_{\alpha, df}$ Reject H_0 if $t \leq -t_{\alpha, df}$ p-value = $\Pr(T < t)$	
Two Tailed test $H_0 : \mu_1 - \mu_2 = D_0$ $H_A : \mu_1 - \mu_2 \neq D_0$	Reject H_0 if either $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$ p-value = $2 * \Pr(Z > z)$	Reject H_0 if either $t \geq t_{\alpha/2, df}$ or $t \leq -t_{\alpha/2, df}$ p-value = $2 * \Pr(T > t)$	

Example: Start-up Owner's Equity

- Two populations
 - Small business firms that have resulted in bankruptcy within the first year of operation
 - Firms that are not bankrupt
- Variable of interest: Owner's equity
- Data

1. Bankrupt firm	11	5	3	1	11	1	3	5	9	8	3	4	7	6	9
2. Nonbankrupt firm	12	9	8	9	6	11	7	10	11	8	5	3	4	7	8

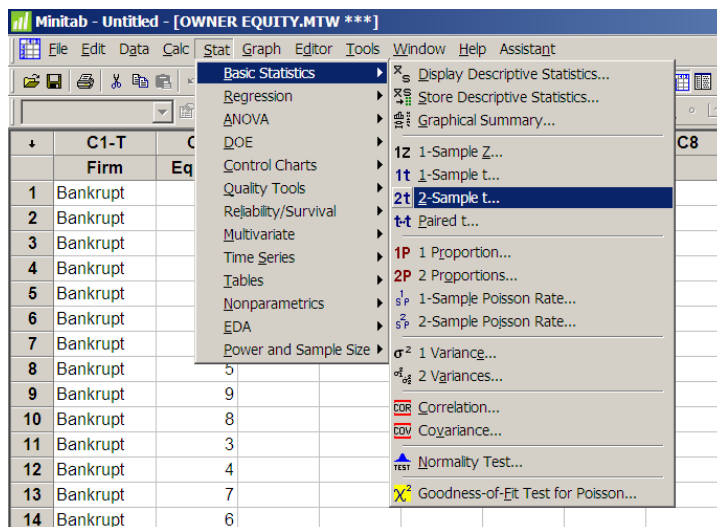
Example: Start-up Owner's Equity

- Is there a difference in the owners' average equity between bankrupt firms and nonbankrupt firms?
- Let μ_1 = Owners' average equity of bankrupt firms
 μ_2 = Owners' average equity of nonbankrupt firms
- Hypotheses

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_a: \mu_1 - \mu_2 \neq 0$$

Minitab Procedure



Output

Two-Sample T-Test and CI: Equity, Firm

Two-sample T for Equity

Firm	N	Mean	StDev	SE Mean
Bankrupt	15	5.73	3.33	0.86
Nonbankrup	15	7.87	2.61	0.68

Difference = mu (Bankrupt) - mu (Nonbankrup)

Estimate for difference: -2.13

95% CI for difference: (-4.38, 0.11)

T-Test of difference = 0 (vs not =): T-Value = -1.95

P-Value = 0.062 DF = 26

Since p-value 0.062 > 0.05, there is not enough evidence to conclude that the average equity between bankrupt and nonbankrupt firms are different at the 95% confidence level.

Topics

- Comparing two population means
 - Matched (paired) samples
 - Independent samples
- Comparing two proportions

Example: Comparing Two Proportions

- During the recent elections survey, the Republican presidential candidate showed the following pre-election voter support in Indiana and Alabama

State	Voters Surveyed	Voters Favoring the Republican Candidate
Indiana	800	440
Alabama	600	360

- We want to determine, with 95% confidence, whether or not the proportions of voters favoring the Republican candidate were the same in both states.

Derive test statistic

$\bar{p}_1$ = sample proportion from population 1, $\bar{p}_2$ = sample proportion from population 2.

We want to test:

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_a : p_1 \neq p_2 \text{ (or one-sided).}$$

1. Sampling distributions of individual sample proportions

For large n_1 and n_2 :

$$\bar{p}_1 \sim N\left(p_1, \frac{\sqrt{p_1(1-p_1)}}{\sqrt{n_1}}\right), \quad \bar{p}_2 \sim N\left(p_2, \frac{\sqrt{p_2(1-p_2)}}{\sqrt{n_2}}\right)$$

Since the samples are independent:

$$\bar{p}_1 - \bar{p}_2 \sim N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

Derive test statistic

2. Pooled proportion and test statistic

Under $H_0 : p_1 = p_2 = p$, define the **pooled sample proportion**:

$$\bar{p} = \frac{n_1 \bar{p}_1 + n_2 \bar{p}_2}{n_1 + n_2}.$$

The **standard deviation** of $\bar{p}_1 - \bar{p}_2$ under H_0 is

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\bar{p}(1-\bar{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}.$$

The **test statistic** is

$$z = \frac{\bar{p}_1 - \bar{p}_2}{\sigma_{\bar{p}_1 - \bar{p}_2}} = \frac{\bar{p}_1 - \bar{p}_2}{\sqrt{\bar{p}(1-\bar{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

Testing for TWO Population Proportions

z-test statistics =

$$\frac{\bar{p}_1 - \bar{p}_2 - 0}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}; \bar{p} = \frac{n_1\bar{p}_1 + n_2\bar{p}_2}{n_1 + n_2}$$

Upper tailed test $H_0: p_1 - p_2 \leq 0$ $H_A: p_1 - p_2 > 0$	Lower tailed Test $H_0: p_1 - p_2 \geq 0$ $H_A: p_1 - p_2 < 0$	Two Tailed test $H_0: p_1 - p_2 = 0$ $H_A: p_1 - p_2 \neq 0$
Reject H_0 if $z \geq z_{\alpha}$ p-value = $\Pr(Z > z)$	Reject H_0 if $z \leq -z_{\alpha}$ p-value = $\Pr(Z < z)$	Reject H_0 if either $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$ p-value = $2 * \Pr(Z > z)$

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Example: Comparing Two Proportions

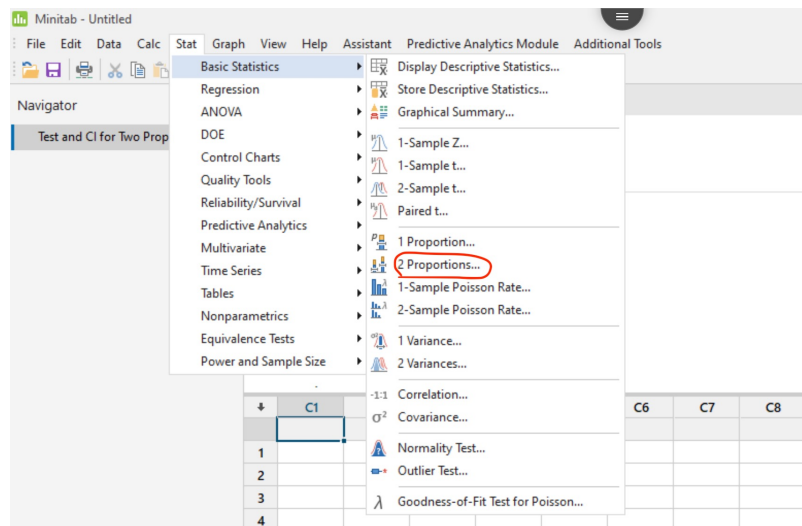
- Let p_1 = proportion of voters favoring the Republican candidate in Indiana
 p_2 = proportion of voters favoring the Republican candidate in Alabama
- Hypotheses $H_0: p_1 = p_2$
 $H_a: p_1 \neq p_2$
- Calculate pooled proportion $\bar{p}$ and the test statistic z
- Determine critical value and the conclusion

EXAMPLES, PRACTICE QUESTIONS

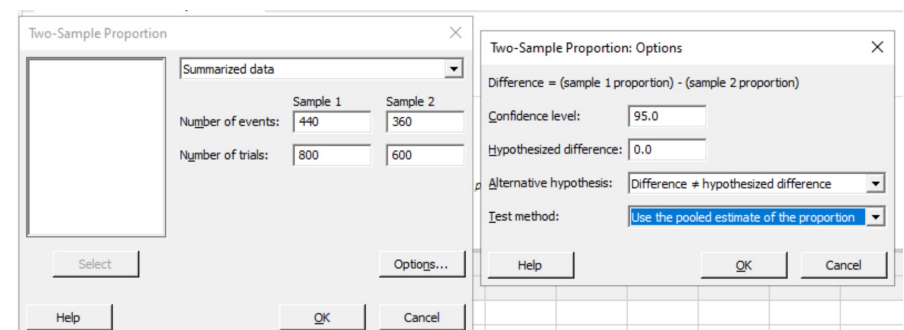
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Minitab Procedure



Minitab Procedure



Test

Null hypothesis $H_0: p_1 - p_2 = 0$
Alternative hypothesis $H_1: p_1 - p_2 \neq 0$

Method	Z-Value	P-Value
Normal approximation	-1.87	0.061

Conclusion: Since p-value > 0.05, at the 5% significance level, we do not have enough evidence to conclude that the proportion of voters favoring the Republican candidate is different between Indiana and Alabama.

Summary

- Comparing two population means
 - Matched (paired) samples
 - Independent samples
- Comparing two proportions
 - z-test

Questions on A/B test are common in DS interview

<https://datalemur.com/blog/ab-testing-interview-questions-and-answers>